

Muneeb ur Rehman

Al-Noor Town, Alipur, Islamabad | muneebqureshi2125@gmail.com | +92 336 0062333

<https://www.linkedin.com/in/muneeb-ur-rehman-1433132125d/>

Objective

Fresh Physics graduate with strong interest in **Computational physics**, **Machine learning** and **Deep learning**. Skilled in using programming and data analysis to model physical system and apply AI in scientific research.

Skills

- **Technical Skills:** ML/DL, Statistical Data Analysis and EDA.
- **Programming Languages:** Python, C.
- **Mathematical Software's:** MS-Excel, Mathematica.
- **Soft Skills:** Communication and Leadership.

Education

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|---|-----------------------|
| Pakistan Institute of Engineering and Applied Sciences, BS-Physics | Sept 2021 – July 2025 |
| • Coursework: Computational Physics, Classical Physics, Quantum Physics, Fundamental of ML, EMT and Solid State Physics. | |
| Punjab Daanish Boys School, DG Khan, FSc (Pre-medical) | Aug 2019 – July 2021 |
| • Course Work: Biology, Physics and Chemistry. | |
| Punjab Daanish Boys School, DG Khan, Matric (Science) | Aug 2017 – July 2019 |
| • Course Work: Math's, Physics, Biology, Chemistry. | |

Projects

Ensemble Machine Learning Approach to Predict the Glass Forming Ability of Bulk Metallic Glasses (FYP)

- Built an ML-based model to predict glass-forming ability of BMGs using simulated and literature-based alloy data. Applied RFE for feature selection, Baysien optimization for model tuning, and Staking approach for ensemble learning.
- **Final Results:** $R^2 = 0.93$, RMSE = 1.42, and MAE = 0.77.
- **Tools Used:** Python(Including Libraries), Google Colab, MS-Excel, HEAP's Software,

Machine Learning Approach to Predict the Yield Strength of Material

- Open source data collection, data processing and train multiple regression model, with RF and Ex-Tree Regression ($R^2=0.97$) gives maximum accuracy.
- **Tools Used:** Python, Google Colab, MS-Excel, Kaggle.

Breast Cancer Detection

- Train Model on Breast Cancer dataset (Images), with obtaining 98% on CNN and SVM neural networks.
- **Tools Used:** Python, Google-Colab, Kaggle.

Experience

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|---|----------------------|
| Machine Learning Intern, EvoSoft – (Remote) | May 2024 – Aug 2024 |
| • Brest Cancer Detection using the Imaging Data Set (MRI and X-Rays Photos), trained using CNN and SVM algorithms, successfully obtained 98 percent accuracy. | |
| Summer Intern, NILOP – Islamabad | June 2024 – Aug 2024 |
| • Syhthesis, fabrication, and characterization of Mg-Amino NPs. | |
| • Synthesis of silver nanoparticles (Ag-NPs), used for "Water Treatment" under UV light. | |

Courses

The Science of Success: What Researchers Know that you Know – <i>University of Michigan</i>	March 2025
Basics of Machine Learning – <i>Great Learning</i>	June 2024
Intro to Exploratory Data Analysis using Excel – <i>Great Learning</i>	Sept 2023
Excel Basics for Data Analysis – <i>IBM</i>	April 2023
Crash Course on Python – <i>Google</i>	March 2023

Additional Experience And Awards

1st International Workshop on Recent Advances in Quantum Science and Technology: – <i>CMS, PIEAS</i> Organized and participated in a 5-day workshop covering topics from basic quantum science to advanced applications like physical proton neural networks in quantum computing.	Feb 17-21, 2025
Managemnet Team's Member – <i>PIEAS Physics Society.</i> An event was arranged to introduce students with classical and quantum nature of universe.	Aug 2022
Letter of Appreciation in Academics – <i>PDBS, DG Khan</i> Letter of Appreciation was received from respected School Principal due to outstanding academic performance in Matric.	Sept 2019